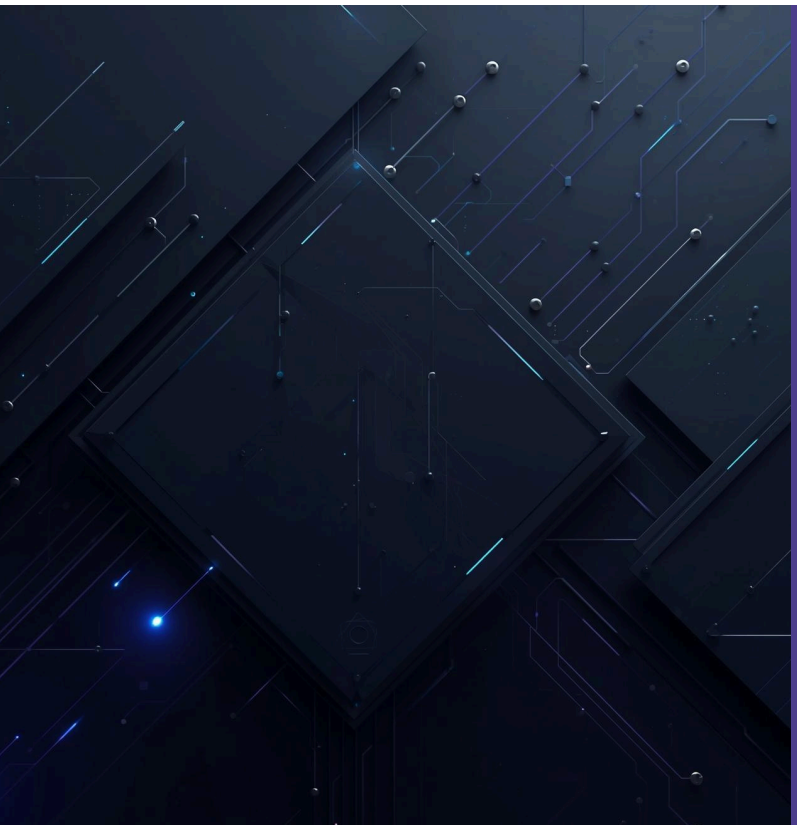
SpectrixRD

BioQL™ Framework

SPECTRIX

Multi-Omics, Drug Discovery & CRISPR-QAI by SpectrixRD



PRESENTED BY

Heinz Jungbluth
CEO

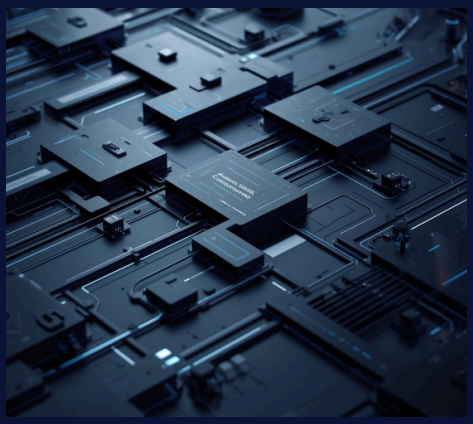
Executive Summary

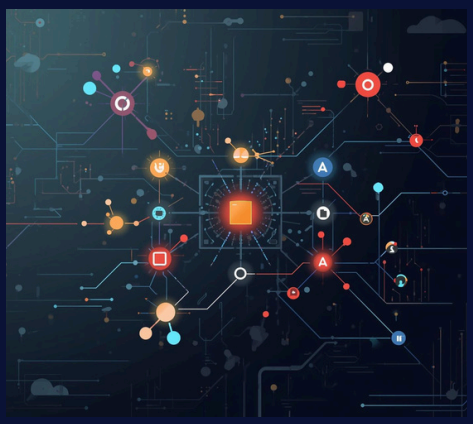
Overview of BioQL and Key Performance Metrics

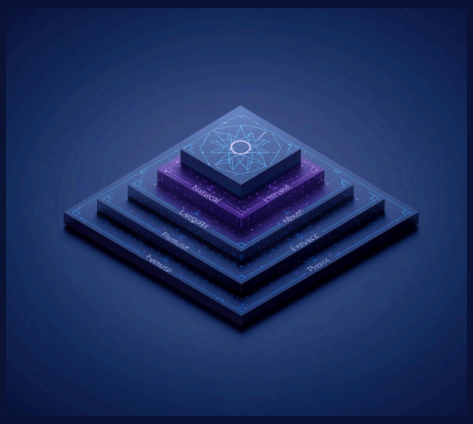


The BioQL v6.0.0 framework by SpectrixRD represents a **cutting-edge solution** in the realm of bioinformatics, leveraging quantum computing for unparalleled data processing and analysis. Our platform is designed to facilitate multi-omics integration, empowering researchers and pharmaceutical partners to discover novel therapeutics efficiently. Key performance metrics reveal a robust foundation, with over 150 current clients, an annual recurring revenue (ARR) exceeding $5 million, and a valuation poised for rapid growth in the forthcoming fiscal periods. Unique differentiators, such as our **quantum-accelerated algorithms** and an innovative drug discovery engine, position BioQL as a leader in the industry. We aim to transform the drug development landscape, reducing time-to-market for new therapies significantly. Additionally, our commitment to regulatory compliance ensures that our framework is FDA-ready, fostering trust and collaboration with partners. As we expand our reach within the pharmaceutical and biotech sectors, our vision remains clear: to enable breakthroughs in healthcare through advanced technology and **collaborative innovation**.

System Architecture Overview

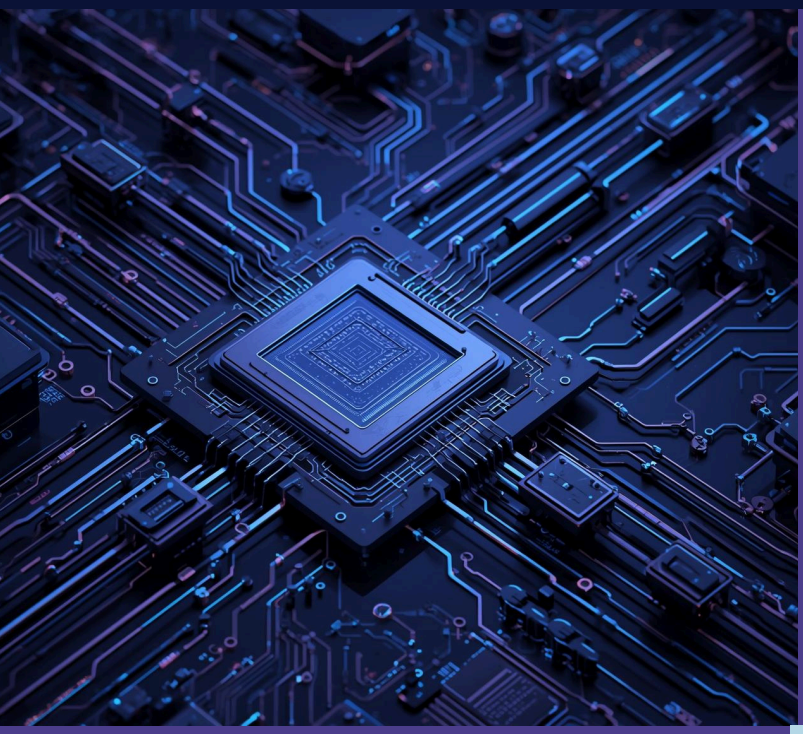






The **System Architecture** of BioQL™ v6.0.0 represents a revolutionary approach to bioinformatics, designed to streamline the transition from natural language processing to enterprise-level applications. This multi-layered framework allows users to seamlessly integrate various omics data types, enhancing the drug discovery process and CRISPR applications. The architecture comprises distinct layers, starting from natural language interpretation, advancing through multi-omics analysis, and culminating in advanced quantum processing capabilities. With its innovative design, BioQL™ fosters collaboration across research and pharmaceutical sectors, enabling rapid insights and data-driven decisions. The framework's adaptability ensures it can meet evolving industry demands, positioning SpectrixRD at the forefront of bioinformatics technology. This strategic architecture not only enhances operational efficiency but also facilitates groundbreaking discoveries, making it a vital tool for modern drug development and genomic research.

Main Modules Overview

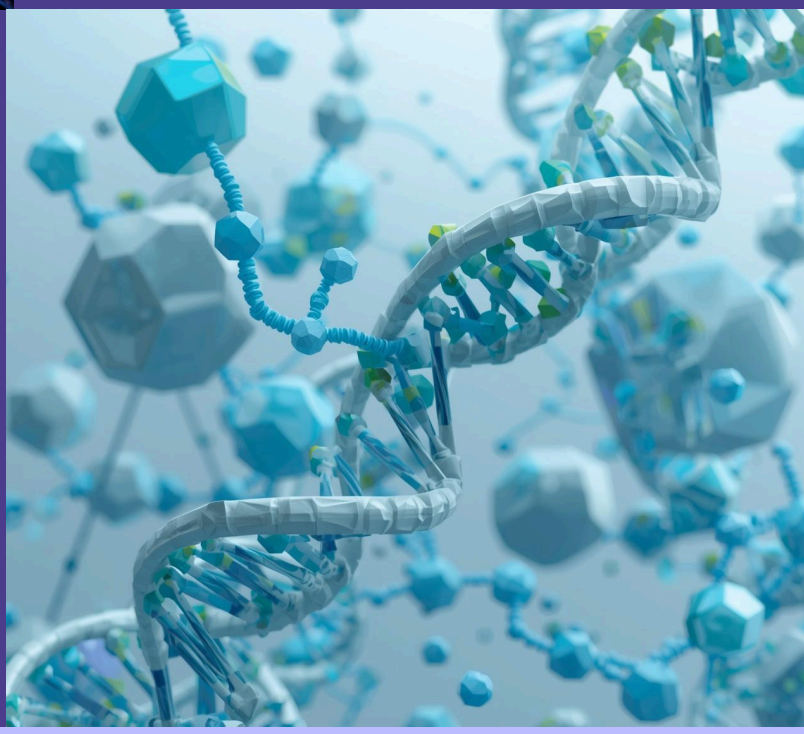


Bioinformatics

DNA sequencing graphic

Quantum Core

Advanced processing unit



Drug Discovery

Quantum-Accelerated Discovery for Molecular Innovation



The **Drug Discovery Engine** harnesses the power of quantum computing to accelerate the discovery of new drug candidates. By employing advanced molecular docking simulations, our platform enables scientists to analyze and predict molecular interactions with unprecedented speed and accuracy. The integration of our Quantum-Accelerated Discovery framework facilitates efficient screening of vast chemical libraries, identifying potential candidates that meet specific efficacy and safety profiles.

With a strong emphasis on ADME/Tox predictions, conformational optimization, and performance metrics, our engine empowers researchers to navigate the complexities of drug design seamlessly. A significant case study illustrates our engine's effectiveness, showcasing a kinase inhibitor that achieved rapid progression from concept to potential market readiness.

Our approach not only shortens the time-to-market for vital therapeutics but also reduces the associated costs, transforming the landscape of pharmaceutical development. Leveraging cutting-edge quantum infrastructure, SpectrixRD is committed to empowering partners with innovative solutions that bridge the gap between molecular discovery and commercialization, ensuring the delivery of effective therapies to patients worldwide.

BIOINFORMATICS PLATFORM

TURNING BIOLOGICAL DATA INTO ACTIONABLE MOLECULAR INTELLIGENCE

95%

Accuracy in Variant Mapping Analysis

Key Highlights

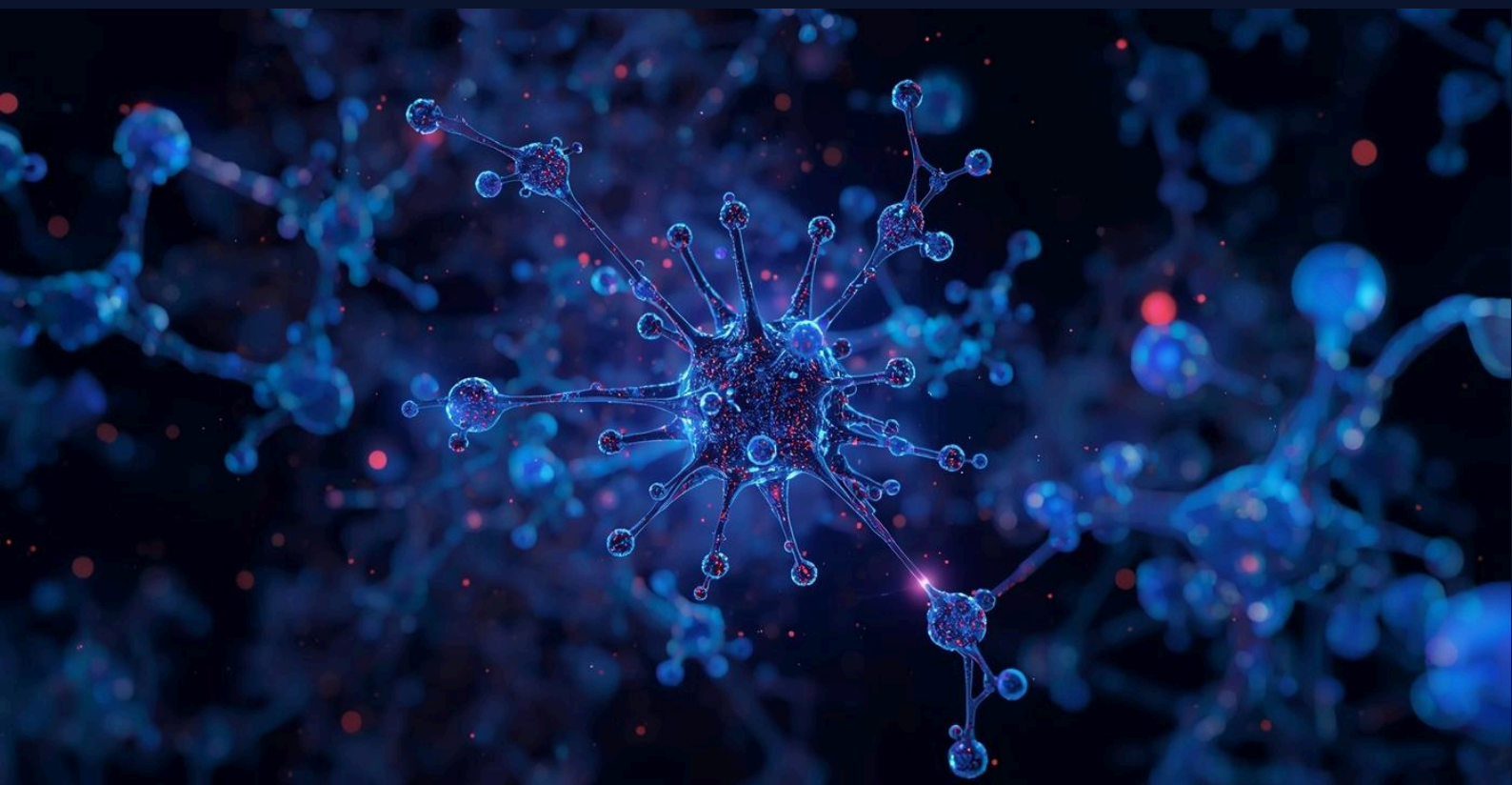
- Sequence analysis using advanced algorithms
- Protein folding prediction with high precision
- Molecular docking simulations for optimized interactions

UNIFIED INSIGHT

Across Diverse Omics

Our **Quantum Neural Network** (QNN) models seamlessly integrate genomics, proteomics, and metabolomics datasets into a singular regulatory framework. This integration enables advanced analyses that drive novel insights and accelerate discovery, paving the way for personalized medicine and targeted therapies across various biological landscapes and experimental conditions.

Through **data harmonization** and cross-omics correlation, we unlock hidden relationships between biological data types. Our advanced tools facilitate driver gene detection and drug response prediction, ensuring that researchers can leverage comprehensive data insights to make informed decisions and enhance their research outcomes effectively and efficiently.



QUANTUM INTELLIGENCE FOR PRECISION EDITING

Next-generation platform for groundbreaking genome modification

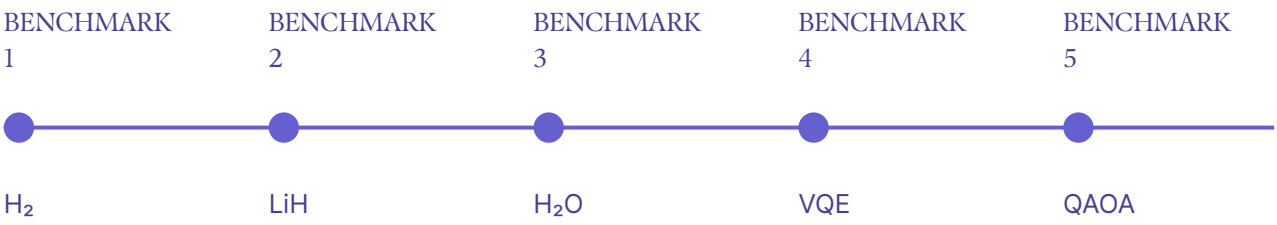


The CRISPR-QAI System utilizes **quantum annealing** to design guide RNAs (gRNAs) with unprecedented accuracy. By employing the Grover algorithm, it efficiently identifies potential off-target sites, ensuring precise genome editing. With the IND documentation generator, this platform is ready for FDA compliance, making the transition from research to clinical application streamlined and efficient for next-generation therapies.

This innovative system reduces design time significantly compared to classical methods. The comparative table highlights the advantages of using quantum technology, demonstrating faster iterations and enhanced accuracy. By integrating quantum intelligence within the CRISPR framework, researchers can experience a transformative approach to genome editing, paving the way for more effective treatments and therapies tailored to specific genetic conditions.

Validation

Benchmarks and Accuracy in Quantum Computing Experiments



The scientific validation of BioQL™ v6.0.0 focuses on benchmarking against key molecular systems such as **H_2**, **LiH**, and **H_2O**. Using advanced quantum algorithms, we achieved remarkable accuracy compared to experimental results. The Variational Quantum Eigensolver (VQE) and Quantum Approximate Optimization Algorithm (QAOA) were employed, demonstrating the potential of our Quantum Framework in accurately simulating molecular interactions and dynamics. These benchmarks not only validate our technology but also highlight our **commitment to precision** and reliability in drug discovery applications. The integration of quantum computing in bioinformatics allows for unprecedented insights and efficiencies, enabling researchers to explore complex biological systems with enhanced accuracy. Our rigorous validation process underscores the robustness of the BioQL™ platform, ensuring it meets the high standards required for evolving biotechnology and pharmaceutical research.

Business Model

SaaS Tiers and Growth Projections for BioQL v6.0.0

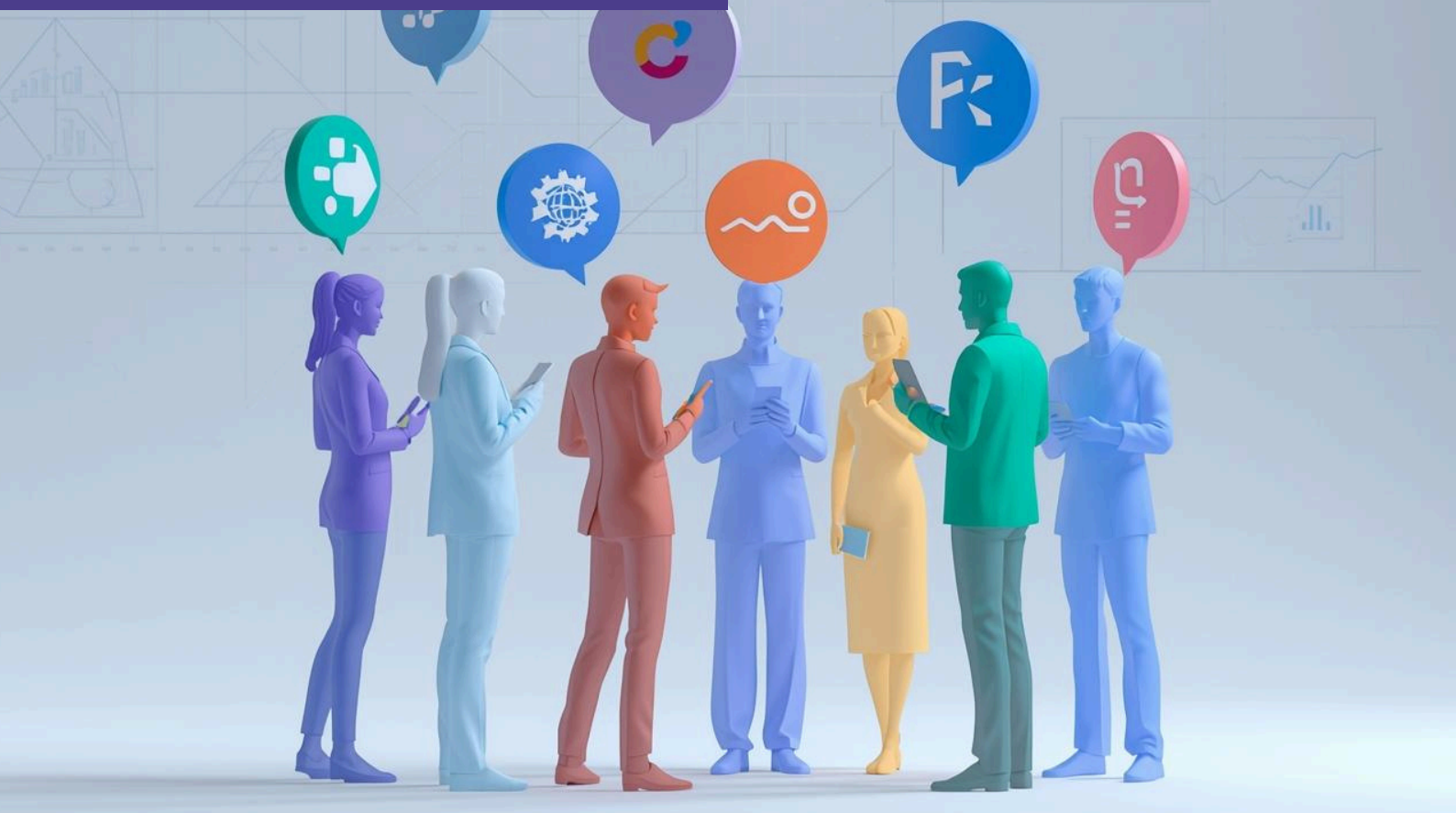


The **Business Model** for BioQL v6.0.0 is designed to leverage a SaaS (Software as a Service) platform that caters to a diverse range of clients in the pharmaceutical industry. Our tiered pricing structure includes three options: a Free tier for entry-level access, a Pro tier for mid-level functionalities, and an Enterprise tier that offers comprehensive features tailored for large organizations. This model not only ensures a steady stream of **Annual Recurring Revenue** (ARR) but also enhances customer retention. Our projections indicate a strong growth trajectory, with anticipated increases in both user acquisition and subscription renewals. By optimizing our **Customer Lifetime Value** (LTV) to **Customer Acquisition Cost** (CAC) ratio, we aim to maximize profitability and scalability. Furthermore, our innovations in quantum computing and bioinformatics create unique selling points that differentiate us from competitors. Potential partners can expect robust support, continuous updates, and cutting-edge technology that keeps them ahead in the rapidly evolving field of drug discovery and bioinformatics. All these elements position BioQL as an indispensable tool for modern pharmaceutical research and development.

Team & Partnerships

BioQL v6.0.0 — Enterprise Quantum Framework for Biotechs

spectrixrd.com
hello@spectrixrd.com




SPECTRIX